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Author(s): Margaret B. Davis, Mark W. Schwartz and Kerry Woods

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Detecting a species limit from pollen in sediments

MARGARET B. DAVIS, MARK W. SCHWARTZ* and KERRY WOODS† *Department of Ecology, Evolution and Behavior, University of Minnesota, Minneapolis, MN 55455, U.S.A., *Illinois Natural History Survey, Champaign, IL 61820, U.S.A., and †Science Division, Bennington College, Bennington, VT 05201, U.S.A.*

Abstract. Pollen percentages in presettlement lake sediments from Michigan, Wisconsin and Minnesota were compared with the range maps of eastern hemlock (*Tsuga canadensis* (L.) Carr.) and American beech (*Fagus grandifolia* Ehrh.) in the early nineteenth century, compiled from the bearing-tree records of the U.S. Public Land Survey. The species limit shown in the land-survey records corresponds within a few kilometres to the limit of living trees. Pollen percentages for both species in presettlement lake sediment fall to low levels 15–20 km beyond the land-survey range limit. Percentages are variable within 10 km of the species limit, and therefore the position of the limit cannot be determined from pollen percentages at a single site. With a grid of fossil sites, however, it is possible to

identify an area 20 km wide that includes the past species limit. On the basis of our data from the Great Lakes region, the past existence of outlying colonies of hemlock and beech can be demonstrated if pollen percentages at a fossil site are consistently higher than at surrounding sites, but it is impossible to prove from pollen that outlying colonies do not exist. Although regression models lead to similar conclusions, information about pollen percentages outside the species limit provided by y-intercept values is less precise than information from a geographical grid of sites.

Key words. Species range limits, land-survey records, pollen percentages, fossil sites, *Fagus grandifolia*, *Tsuga canadensis*, Great Lakes region.

INTRODUCTION

Changes in species range limits through time are of general interest to biogeographers, ecologists and climatologists. Recent papers have discussed range extension and population expansion of forest trees in North America during the Holocene. A number of authors have focussed on regional patterns of migration and their implications for paleoclimate (Davis, 1976, 1981; Davis, Mack & Calcote, 1986; Davis & Jacobson, 1985, 1985; Webb & Bernabo, 1977; Webb, 1986; Gaudreau & Webb, 1985; Delcourt & Delcourt, 1987). Others have emphasized population dynamics, including seed dispersal, competition and rates of population growth (S. L. Webb, 1986, 1987; Davis, 1987; Bennett, 1985, 1988; Woods & Davis, 1989; Tsukada, 1981, 1982; Bennett & Lamb, 1988; Ritchie, 1986). A large literature discusses tree migrations in Europe (Huntley & Birks, 1983; Huntley, 1988). At least one attempt has been made to model past range extensions (Dexter, Banks & Webb, 1987).

All of these studies depend on fossil pollen to provide evidence for the locations of species range limits in the past. It is therefore critical to define the pollen criteria that can be used to detect species presence or absence. These definitions must state clearly the size of the area referred to when 'local' presence is deduced from fossil pollen. When

data are generalized to 1000-year intervals and to regions as large as 10^6 km² (e.g. Davis, 1981; Webb, 1987), such ambiguities have little influence on interpretations. Many recent papers, however, discuss specific colonization events and the spatial structure of the expanding population front. For these more detailed interpretations, the degree of confidence that can be placed on range-limit locations becomes important.

The literature has used three different methods to determine whether a fossil-pollen site was inside or outside the ancient species limit. First, a geographical method compares range maps with pollen percentages in surface samples, with the objective of calibrating pollen percentage values against the species range limit. As a result a critical pollen percentage value is identified (usually 1–3%) that is never exceeded outside the range limit. This value – sometimes called the 'rational limit' – becomes the minimum level that pollen must attain to indicate past species presence in local vegetation (e.g. Davis & Jacobson, 1985; Faegri & Iversen, 1950). A logical problem with the rational limit, even assuming it is correctly determined, is that pollen percentages exceeding the rational limit provide evidence for the *presence* of nearby source plants, whereas percentages below the rational limit are not convincing evidence of the regional *absence* of source plants. For example, low values can be found in surface samples from

unusual habitats well within the geographical range of a species.

The second method, regression, uses models that relate pollen abundances in surface sediments to percentages of trees in the surrounding forest. Regressions often show a positive y-intercept, representing the predicted pollen value when local tree abundance is zero. This is background pollen reaching the site from distant vegetation (Andersen, 1979; Prentice, Berglund & Olsson, 1987). The y-intercept, like the rational limit mentioned above, provides a cutoff point – a pollen value that when exceeded in fossil material is taken as evidence that the source species was locally present (Prentice, 1986). For example, y-intercepts from regressions were used to define the 0-abundance isopleth on the paleo-dominance maps published by Delcourt & Delcourt (1987). The primary problems here are twofold: (a) the y-intercept and the width of its confidence interval are sensitive to the number of sites with no regional source trees that are included in the data set. (b) The regression is scale-dependent, with a different slope and intercept depending on the size of the pollen-collecting site and the size of the vegetation area included in the sample (Bradshaw & Webb, 1985). While Andersen's original use of the concept defined these items very precisely, subsequent applications have seldom given clear definitions to the terms 'local', 'regional' and 'background'.

A third method employs pollen influx. A sharp, 10-fold increase through time in fossil pollen influx was cited as evidence for local 'arrival' of northward-diffusing tree species during the Holocene (Davis, 1981, 1983). Pollen influx rates are seldom calibrated against vegetation, however, because influx is difficult to measure in modern sediments, and because focussing of sediment causes a high variance within and between lakes (Davis & Brubaker, 1973; Davis & Ford, 1982). We have not measured modern pollen influx at most of the sites used in this paper and therefore will not discuss this method further.

The use of critical values determined by geographical correlation, or through regression models, implies a discontinuity between pollen values from distant sources and pollen produced by 'local' trees. Bennett (1988) and Woods & Davis (1989) point out that a sharp cutoff is unlikely. Both authors observe that fossil pollen profiles recording colonization events show increasing pollen influx through time. Woods & Davis consider the earliest pollen occurrences to be wind-blown pollen from distant (>20 km) large (area >1000 km²) populations. Bennett, in contrast, believes the earliest pollen occurrences record expansion of local populations of trees. He argues that because exponential curves can be fitted to pollen influx against time, extending back to the oldest samples in which pollen occurs, the oldest pollen grains are not a constant background blown from distant sources, but instead represent the first occurrence of an expanding population of local trees (Bennett, 1985, 1988).

We agree with Bennett that a sharp cutoff cannot distinguish local from long-distance pollen, but disagree about the mechanism. We would emphasize that the background itself is not constant, because pollen transported from a distant population will increase more or less logarithmical-

ly with decreasing distance from source. As the distant population moves closer to the site, transported pollen will increase. For example, trace quantities of beech pollen are found at all sites in Michigan and Wisconsin in the millennium before the time when large quantities (5–15%) of beech pollen appear (Woods & Davis, 1989). Trace quantities characterized a belt of sites about 100 km wide, which can be seen on maps (Fig. 4, Woods & Davis, 1989), which was gradually displaced westward as the group of sites with large (5–15%) percentages of beech pollen expanded over a larger and larger area in the east. Woods & Davis (1989) interpret these occurrences as windblown pollen from a dense population of trees several 10s of km to the east, although they do not preclude the existence of local populations. Bennett's (1985, 1988) papers imply that he would interpret them as evidence for a geographically extensive but sparse population of beech colonizing new territory.

It is our purpose here to develop criteria for distinguishing windblown pollen from locally produced pollen, taking care to define 'local' very precisely. We seek also to compare the accuracy of the geographical and regression methods for detecting past species limits by examining pollen abundances in lake sediments at varying distances from known geographical range limits. All of our data concern two tree species, American beech (*Fagus grandifolia*, Ehrh.) and eastern hemlock (*Tsuga canadensis* (L.) Carr.). These species are of particular interest for studies of dispersal because beech is animal-dispersed whereas hemlock seeds are dispersed by wind. The beech range has a continuous front, whereas hemlock has a number of outlying populations 50–150 km west and south of the main range limit (Fowells, 1965). Our 127 sites lie in Minnesota, Wisconsin and Michigan, at varying distances from the species limits and from outlying populations. 'Presettlement' fossil pollen deposited in the early nineteenth century before forest clearance is examined because it is analogous to Holocene material. Surface samples are less desirable because open farmland with woodlots, and secondary forests in which hemlock and beech are much reduced in abundance, now surround many of the lakes. The scarcity and discontinuous distributions of these trees near their range limits today makes it difficult to apply the geographical method to surface sediments, and the narrow range of values of tree percentages and pollen percentages in surface sediments reduces the significance of regressions (Schwartz, 1989). We suspect that production and dispersal of pollen may also be affected directly by the open character of the landscape.

In our application of the geographical method we plot presettlement pollen percentages on maps, comparing the pollen with hemlock and beech range maps compiled from records of bearing trees identified in the early nineteenth century by the U.S. Public Land Survey. Because our pollen and forest data are from primeval forest vegetation, our conclusions should be directly applicable to the Holocene palaeorecord of this region (Davis *et al.*, 1986a, b; S. L. Webb, 1987; Davis, 1987; Woods & Davis, 1989; Calcote, 1986).

Regression analysis was applied to a subset of our data, thirty samples from small lakes of uniform size. Pollen per-

centages in presettlement sediment were compared to abundances in the records of the Public Land Survey of the surrounding township; y-intercepts were compared with 'background' levels observed beyond the range limits using the geographical method.

Finally, we discuss our findings with respect to the precision and accuracy with which past range limits for other tree species can be detected from fossil pollen percentages in lake sediments.

METHODS

Tree mapping

Hemlock and beech tree distributions before Euro-American settlement (1825–64) were tabulated for Wisconsin and part of northern Michigan from the U.S. Public Land Office Records. The surveyors used 'bearing' trees to mark section lines in a 1 mile (1.6 km) grid over the landscape. In Wisconsin the surveyors identified four bearing trees at each section corner (i.e. every mile along survey lines), in Michigan two trees were identified at each corner. In both states, two 'line' trees were identified at the midpoint of each section line, as were additional trees in the line of sight. Verbal descriptions of the forest along section lines often mentioned either hemlock or beech; we recorded all these instances as well as bearing and line trees. In Michigan we inspected records from both inside and outside section lines in all townships (= 36 miles², 92.2 km²) along the beech boundary; in Wisconsin we used only the inside township line records.

Hemlock and beech abundances were based on percentages of total bearing and line trees recorded within each township. On average 165 (standard error = 32) trees of all species were recorded per township, or about 2 trees per km². Although this census is low in density, it is remarkably continuous across the landscape; no modern mapping effort, except remote sensing, provides such extensive coverage. Because hemlock trees are very distinctive, it seems unlikely that surveyors would have overlooked or misidentified them. Beech is also distinctive, although it may have been confused with 'blue beech' (*Carpinus carolinia* Walter). Surveyors may have preferred beech as a bearing tree because records could be written directly on the bark, eliminating the need to cut a blaze. The validity of witness trees as a record of vegetation has been discussed by Bourdo (1956) and Grimm (1981, 1984); most of the problems they recognize are not important in establishing range limits.

Pollen percentages

The uppermost meter of sediment was collected at fifty-eight sites by use of a 1 m long 2.5 cm diameter plastic tube equipped with a piston. In the laboratory, sediment was extruded in 0.5 or 1 cm segments. Laboratory procedures are described by Woods & Davis (1989). The presettlement stratum (i.e. contemporaneous with or slightly predating the Land Survey) was identified from the pollen stratigraphy. Pollen from ragweed (*Ambrosia*) and other agricultural weeds became abundant 10–50 cm below the surface in all

cores while pollen from white pine (*Pinus strobus* L.) and hemlock decreased. These changes are associated with logging and Euro-American settlement (Brubaker, 1975; Gajewski, Winkler & Swain, 1985). Settlement occurred in the mid- to late 1800s in Wisconsin and lower Michigan and in the early 1900s in northern Michigan (Flader, 1983). Sediment 1–5 cm immediately below the *Ambrosia* rise was considered 'presettlement' in age. In most lakes the extrapolated sediment accumulation rate was 10–20 years/cm. Pollen percentages were calculated based on total terrestrial pollen and total tree pollen. At least 300 tree pollen grains were counted per sample.

The data from fifty-eight sediment cores were supplemented by thirty-eight presettlement samples previously analysed in our laboratory (T. Webb III, 1973; Brubaker, 1975; Kerfoot, 1974; S. L. Webb, 1987 and unpubl.; Calcote, 1986; Middeldorp, unpubl.) and by thirty-one analyses from the literature (see Table 1).

Presettlement pollen assemblages from thirty of the Wisconsin lakes were selected for regression analysis. The lakes selected are similar in size (3–10 ha) and therefore receive similar ratios of pollen from distant versus local sources (Tauber, 1967); locations of sites are shown in Schwartz (1989).

RESULTS

Presettlement distributions of trees

The hemlock and beech distributions (Figs. 1A and 1B) differ slightly from the more generalized distribution maps of Fowells (1965), which indicate a tongue of hemlock distribution extending southward into the driftless region of Wisconsin. The Land Survey shows that the populations in the driftless region were true outliers separated from the main population by almost 100 km. Similarly, hemlock did not extend as far west along the shore of Lake Superior as indicated by Fowells. Fowells (1965) indicates several outliers in Minnesota, 50–100 km beyond the range limit in Wisconsin. The Minnesota Natural Heritage Program identified fifteen hemlock stands in Minnesota; several stands consist of only one or two trees (Calcote, 1986). Survey records in Minnesota record only two occurrences of hemlock (E. J. Cushing, E. Grimm and J. Almendinger, personal communication).

In second-growth forests in northern Wisconsin hemlock occurrence is now patchy and discontinuous, especially along its western and southwestern limit. At the time of the Land Survey, however, hemlock comprised up to 45% of the bearing trees in townships within 10 km of its western and southern limits (Fig. 1B). Logging and fires have apparently eliminated hemlock from large areas and have isolated surviving stands that were once within the continuous range of the species. Regeneration has been inhibited by increased browsing by deer, although decreased seed availability and changed soils and climate may also have played a part (Anderson & Loucks, 1971; Frelich & Lorimer, 1985; Rogers, 1978).

The species limit for beech in presettlement time was more irregular than indicated by Fowells' (1965) map, with

TABLE 1. List of lakes used for the analysis, percentages of hemlock (*Tsuga canadensis*) and beech (*Fagus grandifolia*) pollen, and distances from range limits. Negative distances are inside range limits, positive outside. Pollen percentages from presettlement sediment are calculated as per cent arboreal pollen.

No.	Site name	State	County	Lat.	Long.	Area (ha)	km from hemlock limit	km from beech limit	Depth settlement horizon (cm)	%Hemlock	%Beech	Reference
1	Moss	MI	Delta	45°55'	86°38'	350	-180	-23	27	0.0	1.1	**
2	Whitmore	MI	Livingston	42°27'	83°45'	270	95	-285	*	0.4	2.6	T. Webb, 1973
3	Devil's	WI	Sauk	43°25'	89°44'	152	100	95	22.5	0.2	0.0	Maier, 1982
4	Esau	MI	Presque Isle	45°19'	83°29'	121	-125	-120	*	5.7	2.4	T. Webb, 1973
5	Hudson	IN	Laforte	41°67'	80°53'	93	20	-45	74	0.3	10.0	Bailey, 1972
6	Rifle	MI	Ogemaw	44°25'	83°58'	74	-125	-220	*	6.7	1.5	T. Webb, 1973
7	Duck	MI	Muskegon	43°21'	86°50'	73	-47	-160	*	4.1	4.1	T. Webb, 1973
8	Budd	MI	Clare	44°10'	84°48'	71	-72	-260	*	5.6	2.6	T. Webb, 1973
9	Sunday	MI	Gogebic	46°29'	89°56'	65	-88	150	14	3.3	0.0	**
10	Bryant	MI	Genesee	42°48'	83°48'	53	55	-300	*	1.0	10.3	T. Webb, 1973
11	Clear	MI	Mecosta	43°41'	85°24'	53	-45	-210	*	3.6	3.6	T. Webb, 1973
12	Crystal	MI	Oceana	43°59'	86°15'	51	-103	-140	*	13.6	14.2	T. Webb, 1973
13	Rush	MI	Van Buren	42°14'	86°11'	48	5	-105	*	2.5	25.9	T. Webb, 1973
14	Lee	MI	Calhoun	42°11'	85°70'	47	88	-170	*	0.6	6.4	T. Webb, 1973
15	Crooked	MI	Berrien	42°05'	86°11'	45	15	-90	*	0.0	13.3	T. Webb, 1973
16	Lake 27	MI	Ostego	45°64'	84°47'	45	-172	-172	2	18.9	6.0	T. Webb, 1981
17	Shupac	MI	Crawford	44°49'	84°25'	44	-165	-190	*	6.1	4.0	T. Webb, 1973
18	Silver	MI	Dickinson	46°12'	88°10'	39	-180	37	35	5.1	0.3	**
19	Holland	MI	Montcalm	43°14'	85°30'	32	10	-225	*	2.2	6.5	T. Webb, 1973
20	Marion	MI	Charlevoix	45°14'	85°15'	31	-178	-157	2	26.3	16.5	Bernabo, 1981
21	Otter	MI	Genesee	43°13'	83°28'	28	8	-345	*	2.9	5.8	T. Webb, 1973
22	Pine	MI	Barry	42°35'	85°25'	27	50	-180	*	1.0	5.7	T. Webb, 1973
23	Wood	WI	Taylor	45°20'	90°50'	25.3	-53	78	29	21.0	0.0	Heide, 1984
24	Blind	MI	Livingston	42°26'	84°01'	25	100	-265	*	0.6	3.5	T. Webb, 1973
25	L. of Clouds	MI	Ontonagon	46°48'	89°45'	25	-107	155	18	14.5	0.3	**
26	Manganese	MI	Keweenaw	47°27'	87°51'	24	-230	116	14	1.4	0.3	**
27	Hanburg	MI	Dickinson	45°48'	87°53'	23	-132	0	55	5.6	0.4	**
28	Six mile	MI	Ontonagon	46°46'	88°56'	23	-165	110	45	10.0	0.0	**
29	Gibson	MI	Iron	46°11'	88°26'	22	-183	42	35	5.6	0.3	**
30	Lancaster	MI	Emmet	45°37'	84°42'	20.7	-125	-125	*	14.1	9.6	T. Webb, 1973
31	Clear	IN	Laforte	41°65'	86°53'	20	20	-20	115	0.3	9.1	**
32	Minerva	MI	Schoolcraft	46°08'	82°25'	18	-163	-26	120	11.3	11.3	**
33	Healy	MI	Manistee	44°26'	80°0'	18	-127	-165	*	24.2	11.0	T. Webb, 1973
34	Jones	MI	Crawford	44°46'	84°36'	17	-155	-235	3	7.6	1.5	Bernabo, 1981
35	Lake Mary	WI	Vilas	46°15'	85°54'	16.9	-100	-133	0	12.9	0.0	T. Webb, 1974
36	Green	MI	Antrim	44°53'	85°07'	16	-170	-190	10	6.3	17.0	Lawrenz, 1975
37	Perch	MI	Luce	46°30'	85°36'	15.3	-108	-112	18	5.6	3.1	**
38	Lk of Clouds	MN	Lake	48°0'	91°07'	13.5	75	305	208	0.0	0.0	Craig, 1972
39	Rutz	MN	Carver	44°52'	93°52'	13	200	430	100	0.4	0.0	Waddington, 1976
40	Sun	MI	Gogebic	46°10'	89°04'	12	-165	78	18	8.7	0.0	**
41	McDonald	MI	Delta	46°20'	86°48'	12	-180	-8	75	6.3	1.0	**
42	Farmer	MI	Cheboygan	46°17'	87°20'	11	-195	5	44	5.3	0.4	**

43	Little Silver	WI	Barron	45°35'	91°56'	10.8	31	220	15	0.3	0.3	**
44	Kingburg	MN	St. Louis	46°52'	92°45'	9	48	345	35	0.3	0.0	Calcote, 1986
45	Green	WI	Oconto	45°12'	88°25'	8.9	-65	5	96	9.2	0.4	**
46	Huiras	WI	Sheboygan	43°30'	87°58'	8.6	65	-25	50	0.7	4.8	S. Webb, 1983
47	Stewarts DK.	WI	Bayfield	45°18'	91°27'	8.5	5	178	20	1.7	0.0	Heide, 1984
48	Spruce	WI	Taylor	45°08'	90°40'	8	-20	-110	59	17.6	0.0	Overpeck, 1986
49	Soldier	WI	Chippewa	46°20'	84°52'	7.5	-80	-83	18	2.7	1.7	**
50	Camp 3	WI	Forest	45°37'	88°36'	7.3	-112	15	40	10.4	0.3	**
51	S. Bass	MI	Marquette	46°41'	87°38'	7	-235	45	10	13.7	0.2	**
52	Round	WI	Calumet	44°10'	88°10'	6.8	-13	-15	24	2.0	5.5	S. Webb, 1983
53	Brampton	MI	Delta	45°55'	87°01'	6.5	-166	-6	8	7.7	1.1	**
54	Casey	WI	Waupaca	44°50'	89°50'	6.5	3	8	65	4.4	2.2	**
55	Little Brevort	MI	Mackinac	46°10'	85°0'	6.1	-105	-105	52.5	4.2	4.8	Futyma, 1981
56	Spirit	MI	Alger	46°28'	86°15'	6	-180	-8	12	19.5	2.5	Woods & Davis, 1988
57	Frains	MI	Washtenaw	42°20'	83°25'	6	102	-290	350	1.3	5.3	Kerfoot, 1974
58	B. Bass	MI	Schoolcraft	46°10'	86°28'	6	-202	-31	65	13.4	6.4	**
59	Kellner's	WI	Manitowoc	44°15'	87°45'	6	-3	-33	60	1.1	7.1	Goodwin, 1977
60	Ashfork	MI	Schoolcraft	46°08'	86°18'	6	-157	-43	29	9.8	9.8	**
61	Swift	WI	Walworth	42°49'	88°31'	5.4	148	37	69	0.7	0.7	S. Webb, 1983
62	Taylor	MI	Allegan	42°35'	85°42'	5	30	-165	*	0.9	23.2	T. Webb, 1973
63	Kitchner	MI	Menominee	45°40'	87°27'	5	-130	-25	50	16.9	0.6	Woods & Davis, 1988
64	Ackerman	MI	Alger	46°19'	86°47'	5	-165	-26	30	10.2	13.8	**
65	Brandt	MI	Menominee	45°40'	87°27'	5	-128	-5	30	13.4	0.9	**
66	Planet	WI	Price	45°53'	90°22'	4.9	-60	133	20	18.1	0.0	**
67	E. Soldier	MI	Chippewa	46°20'	84°51'	4.8	-76	-125	15	3.2	2.0	Futyma, 1981
68	Tamarack	MI	Leelanau	45°03'	80°0'	4.5	-185	-80	80	7.6	26.6	**
69	Seidel	WI	Kewaunee	44°27'	87°25'	4.5	-33	-60	30	34.7	1.3	West, 1961
70	Schisel	WI	Manitowoc	44°07'	87°48'	4.4	2	-24	30	2.1	3.1	S. Webb, 1983
71	Kroening	WI	Shawano	44°49'	88°52'	4.4	-28	-10	20	18.0	1.7	**
72	Deep	WI	Lincoln	45°26'	89°28'	4.3	-90	-55	40	12.0	0.0	**
73	Lorraine	MI	Schoolcraft	46°08'	86°28'	4	-163	-26	120	24.6	4.9	**
74	Little Dollar	MI	Mackinac	46°08'	85°18'	4	-116	-117	67	4.7	4.6	Futyma, 1981
75	Bohnes	WI	Marathon	44°56'	89°15'	4	-32	5	70	13.0	0.3	**
76	Hoglund	MI	Menominee	45°28'	87°29'	4	-115	-30	20	6.6	5.9	**
77	Penegor	MI	Houghton	46°37'	88°51'	4	-180	105	13	9.4	0.0	**
78	Weber	MN	Lake	47°28'	91°39'	4	50	280	60	0.6	0	Fries, 1962
79	Beaver	WI	Oconto	44°55'	88°09'	3.6	-32	-25	22	24.1	2.8	**
80	Camp	WI	Bayfield	46°15'	91°20'	3.6	-10	220	50	2.4	0.0	**
81	Kerstan	WI	Shawano	44°42'	89°0'	3.6	-10	1	29	1.6	1.4	**
82	Balsam	WI	Sawyer	45°40'	91°15'	3.5	-17	172	24	3.1	0.3	**
83	Dickman	WI	Fondulac	43°35'	88°10'	3.4	-67	-10	83	0.3	2.7	S. Webb, 1983
84	Martin Pond	MN	Hubbard	47°11'	94°56'	3.3	255	485	20	0	0	McAndrews, 1966
85	Wolverine	MI	Luce	46°25'	85°39'	3.2	-100	-96	23	0.4	0.6	Futyma, 1981
86	Ondris Pond	MN	Cass	46°21'	94°33'	3.2	183	413	34	0.5	0	Jacobson, 1979
87	B. Indian	MI	Dickinson	45°59'	87°50'	3	-148	2	40	13.8	0.3	**
88	Abraham	MI	Marquette	46°22'	87°13'	3	-200	1	10	11.2	0.4	**
89	Seneca	MI	Keweenaw	47°18'	88°22'	3	-220	130	18	3.8	0.0	**
90	Morrison	MI	Iron	46°9'	88°37'	3	-181	50	20	19.2	0.3	**
91	Camp II	MI	Marquette	46°4'	88°20'	3	-225	58	10	3.1	0.0	Brubaker, 1975
92	Jacobson	MN	Pine	46°25'	92°43'	3	73	303	20	0	0	Wright & Watts, 1969

TABLE 1 (continued).

No.	Site name	State	County	Lat.	Long.	Area (ha)	km from hemlock limit	km from beech limit	Depth settlement horizon (cm)	%Hemlock	%Beech	Reference
93	Canyon	MI	Marquette	46°5'	87°55'	3	-235	64	17	23.0	0.0	**
94	Midget	WI	Oconto	45°1'	89°35'	2.8	-65	5	35	10.5	1.3	**
95	Radtko	WI	Washington	43°28'	88°50'	2.8	78	-15	64	0.0	1.1	S. Webb, 1987
96	Gass	WI	Manitowoc	44°30'	87°43'	2.8	8	-32	94	3.0	7.7	S. Webb, 1987
97	Mud	WI	Taylor	45°10'	90°50'	2.8	-5	130	30	14.3	0.0	**
98	Silver	WI	Waupaca	44°25'	89°20'	2.6	8	35	50	2.0	0.4	**
99	Billy's Lake	MN	Morrison	46°16'	94°33'	2.5	190	420	50	0	0	Jacobson, 1979
100	Young	MI	Luce	46°26'	85°43'	2.5	-102	-95	30	7.9	5.5	**
101	Pine	WI	Oconto	45°11'	88°40'	2	-70	1	25	18.0	0.3	**
102	Rahmel	WI	Taylor	45°13'	90°35'	2	-25	115	20	22.2	0.0	**
103	Sodon	MI	Oakland	42°35'	83°18'	2	75	-325	*	2.0	4.5	T. Webb, 1973
104	Gilbert	WI	Douglas	46°40'	91°40'	1.8	18	255	40	0.4	0.0	**
105	Capitola	WI	Lincoln	45°30'	89°40'	1.7	-90	75	15	11.9	0.0	**
106	Saxton	WI	Rusk	45°20'	91°15'	1.6	3	162	21	1.4	0.0	**
107	Kellys	WI	Taylor	45°18'	90°21'	1.6	-41	97	21	24.3	0.0	Heide, 1984
108	Hemlock	WI	Bayfield	46°15'	91°15'	1.5	-10	215	35	3.5	0.0	**
109	Pumpkin Pie	WI	Washburn	45°35'	91°40'	1.5	7	200	30	1.4	0.0	**
110	Goose	WI	Sawyer	45°55'	91°10'	1.5	-6	180	7	6.4	0.3	**
111	Moldahamer	WI	Ozaukee	43°52'	87°56'	1.3	85	-20	75	0.7	18.3	S. Webb, 1983
112	Mory Pond	WI	Chippewa	45°10'	91°15'	1.1	4	160	25	0.5	0.5	**
113	Buck	MI	Alger	46°32'	86°20'	1	-155	-57	83	9.0	2.7	**
114	One	MI	Oscego	45°30'	84°42'	1	-165	-160	*	2.3	6.9	T. Webb, 1973
115	Vestaburg	MI	Montcalm	43°26'	84°52'	0.5	-5	-250	18	4.5	6.4	Gilliam <i>et al.</i> , 1967
116	Wintergreen	MI	Barry	42°25'	85°23'	0.5	60	-170	68	0.0	4.6	Manny <i>et al.</i> , 1978
117	Blue Mounds	WI	Dane	43°50'	85°31'	***	140	80	*	0.0	0.0	A. Davis, 1977
118	Stockton Bog	WI	Bayfield	46°55'	90°35'	***	-42	225	50	0.5	0.0	Swain & Winkler, 1983
119	Brander Bog	WI	Bayfield	46°56'	90°35'	***	-43	225	35	11.8	0.0	Swain & Winkler, 1983
120	Vollo Bog	IL	Lake	42°21'	88°11'	***	125	47	20	0.8	1.4	King, 1981
121	Hody Bog	WI	Door	44°51'	87°19'	***	-40	-80	29	12.6	16.7	**
122	Jenny Lk. Bog	MI	Alger	46°15'	86°49'	***	-210	-28	32	8.7	2.8	**
123	Scotts Bog	MI	Schoolcraft	46°17'	86°32'	***	-210	-30	12	8.5	4.1	**
124	Bog D	MN	Hubbard	47°11'	95°10'	***	270	500	50	0	0	McAndrews, 1955
125	Tamarack Bog	WI	Trempealeau	49°09'	91°27'	***	84	186	*	0.0	0.0	A. Davis, 1979
126	Crystal	MI	Montcalm	43°15'	84°55'	***	15	-250	15	2.0	4.4	T. Webb, 1973
127	Hughs Lk. Bog	MI	Schoolcraft	46°10'	86°28'	***	-163	-26	25	16.0	10.3	**

* = Depth data not available. ** = M. Davis and others, unpublished data. *** = Bog or swamp sediments.

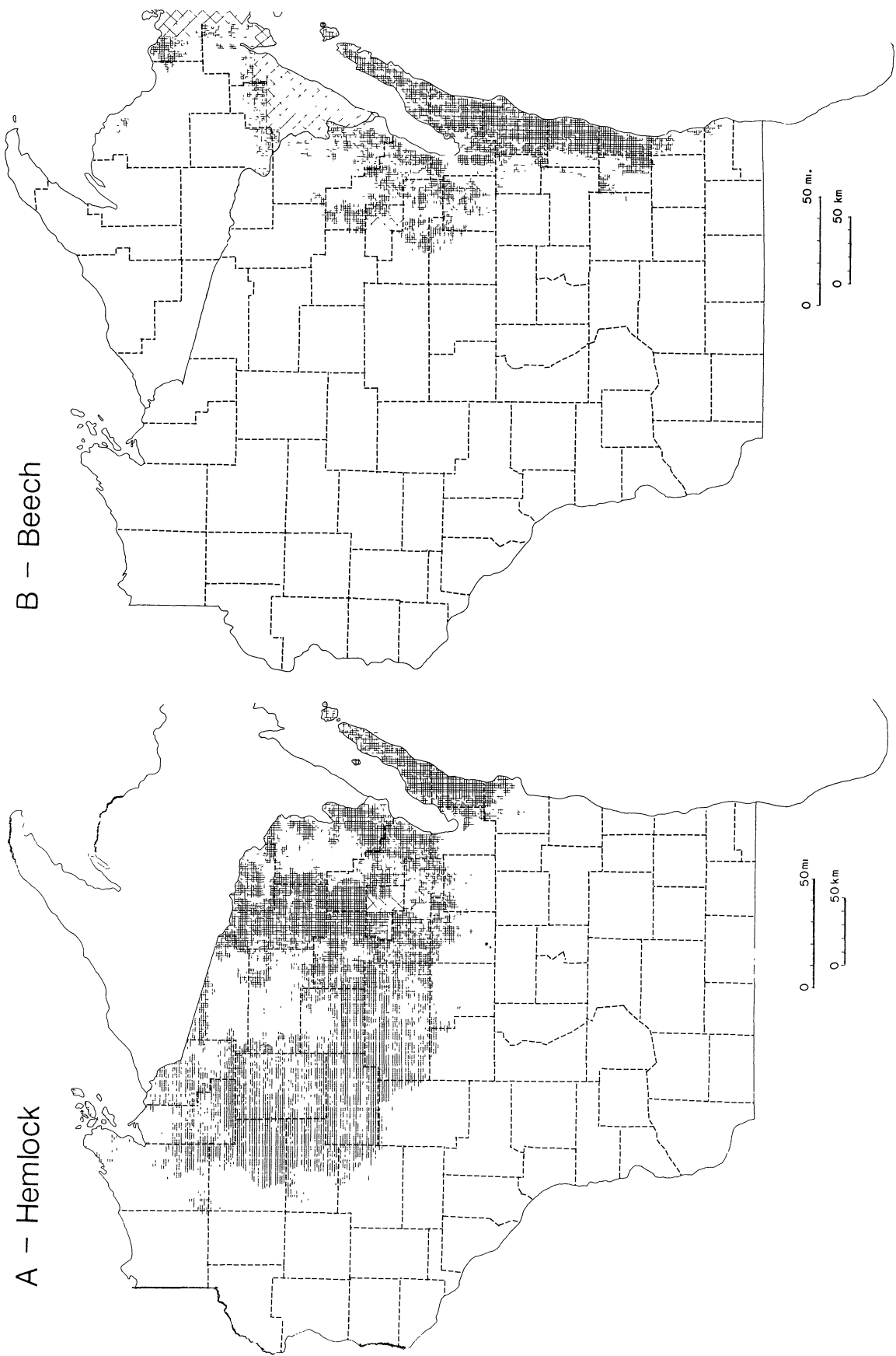


FIG. 1. Maps of hemlock (*Tsuga canadensis*) and beech (*Fagus grandifolia*) based on the occurrences of bearing trees and on forest descriptions in the Land Office Survey Notes. Section lines are shown where hemlock or beech was present.

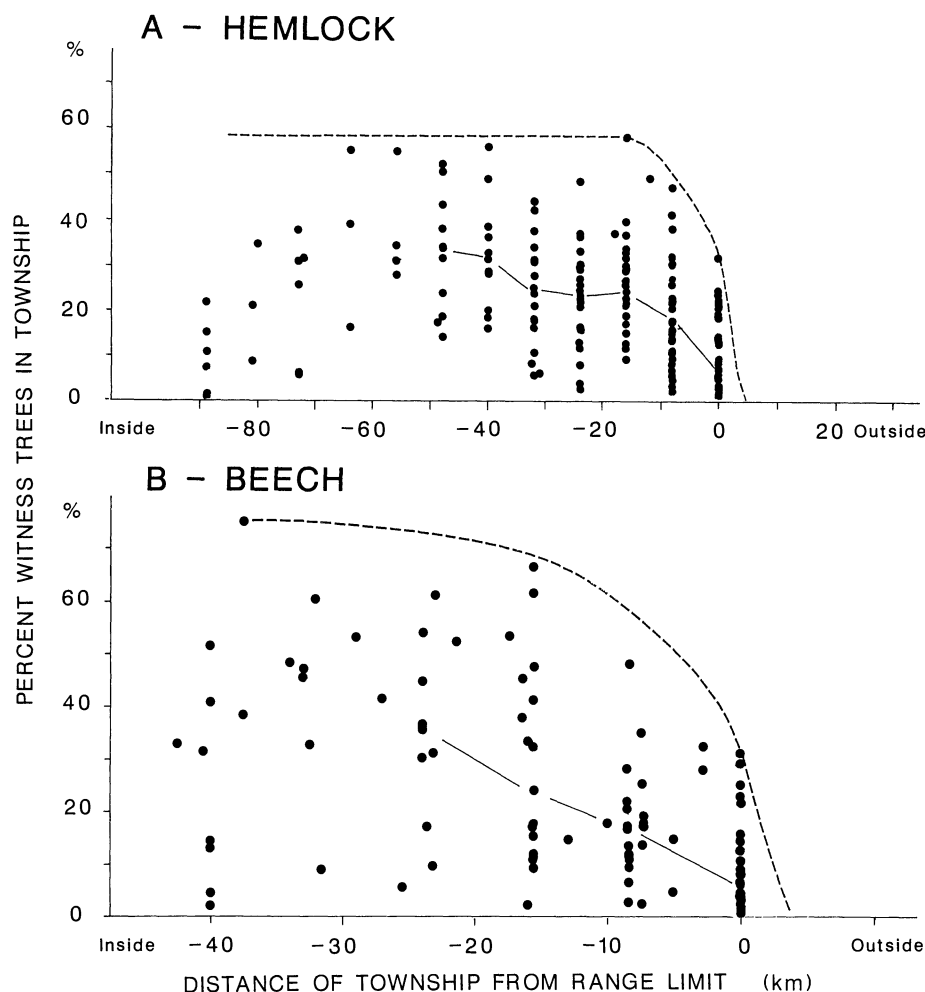


FIG. 2. Abundance of hemlock (*Tsuga canadensis*) and beech (*Fagus grandifolia*) (as per cent witness trees tabulated per township) plotted against the distance of townships from the range limits in Wisconsin and northern Michigan. The range limit was generalized to the outer limit of townships in which the species was reported; as a consequence the townships appear in ranks or tiers depending on their position as a border township, or 1, 2, . . . , n townships back from the limit. Outlying populations are not included.

several outlying populations several kilometres beyond the limit of the continuous range (Fig. 1B). The largest of these stands was 56 km beyond the continuous boundary, north-west of Marquette, Michigan. In northern Michigan near Lake Superior, beech grew along half the section lines in townships right at its western limit. Beech is very common in the second-growth forests that grow there today (Woods & Davis, 1989). In northern Wisconsin the beech boundary followed the limit of glacial drift deposited by the Green Bay and Michigan lobes (Attig & Clayton, 1985). Beech was especially abundant, occurring near most section lines, on soils overlying the Niagara limestone on the Door Peninsula.

We have assessed the accuracy of maps of beech distribution based on Public Land Office records by comparing the locations of beech bearing and line trees with the distributions of living beech trees. Our test area was the township that includes the Dukes Experimental Forest (R.23 W, T.46 N, Marquette County, MI) (Fig. 3). Trees were tabulated along an east-west transect across the township. Near

the eastern boundary of the township, forests were dominated by beech. The western limit of these forests lay 2 km from the eastern end of the transect, at the same location as the westernmost beech trees recorded in the Survey Notes. However, we observed scattered living trees along our transect several kilometres west of this limit. Tree-ring counts indicated that many of these trees were too young to have been recorded at the time of the survey. A number of large and presumably older beech trees were also observed near Abraham Lake (Fig. 3). They were not recorded in the Survey Notes, perhaps because they were too sparsely distributed to occur on section lines or near corners. No beech trees were observed west of the township line by us or by the Public Land Survey.

These results suggest that the species limit derived by Land Survey Notes corresponds quite well with the present limit of abundant beech in the forests. However, the ultimate limit of trees now lies several kilometres farther west than the westernmost trees recorded. In order to include this limit, we generalized all maps of the species range to the

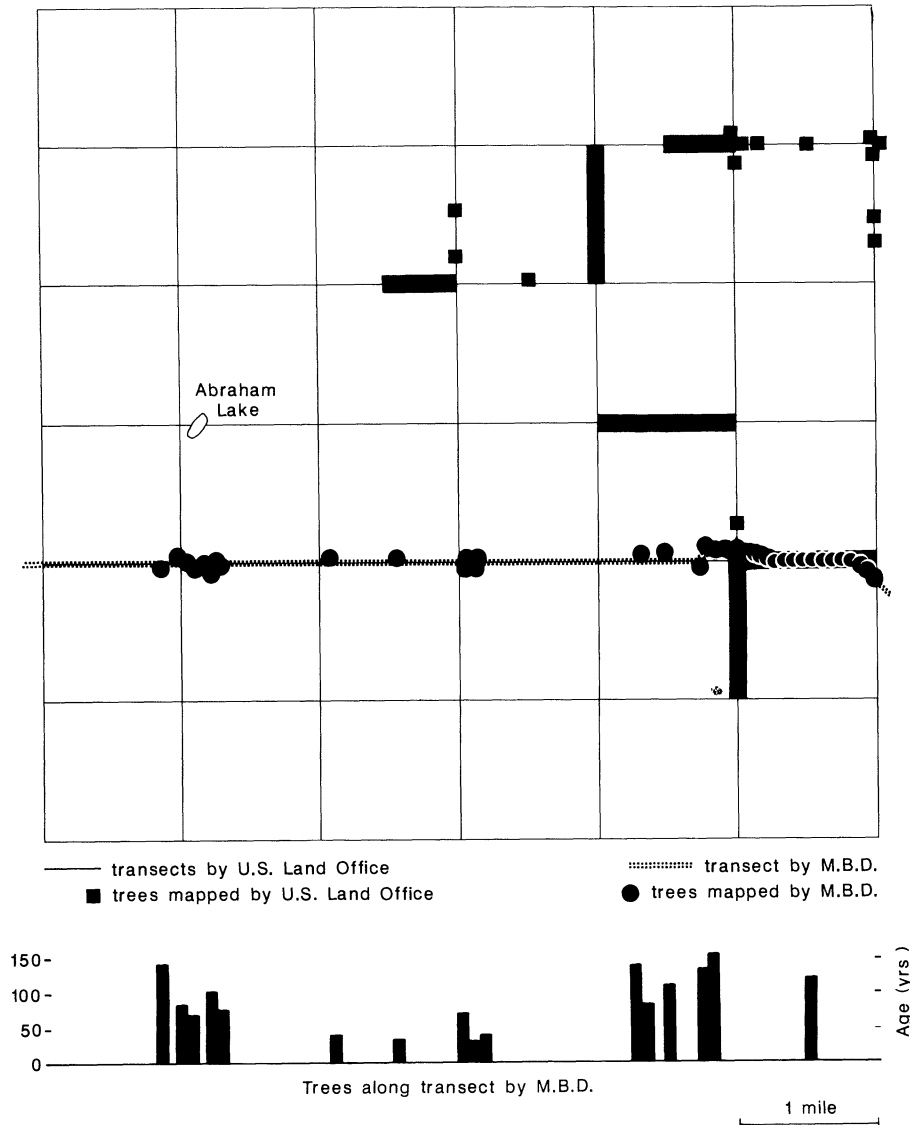


FIG. 3. Map of township in which Dukes Forest is located. Squares indicate trees reported in the nineteenth century by the Land Survey. Dots indicate living trees encountered along a west-east transect across the township. Estimated ages of trees on transect indicated at the bottom of the figure.

outermost boundary of all townships where any beech was reported – a limit that in all cases is no more than 10 km beyond the westernmost trees recorded in the survey. Outlying populations, separated by one or more townships from the main population, are similarly generalized to township. The western, southern and northern boundaries of the township in Fig. 3 are the location of this generalized limit; note the generalized limit includes all the living beech trees that we observed along the transect. Fig. 4 shows generalized beech and hemlock limits at the time of the Public Land Survey, drawn at the outer borders of townships where beech or hemlock were reported.

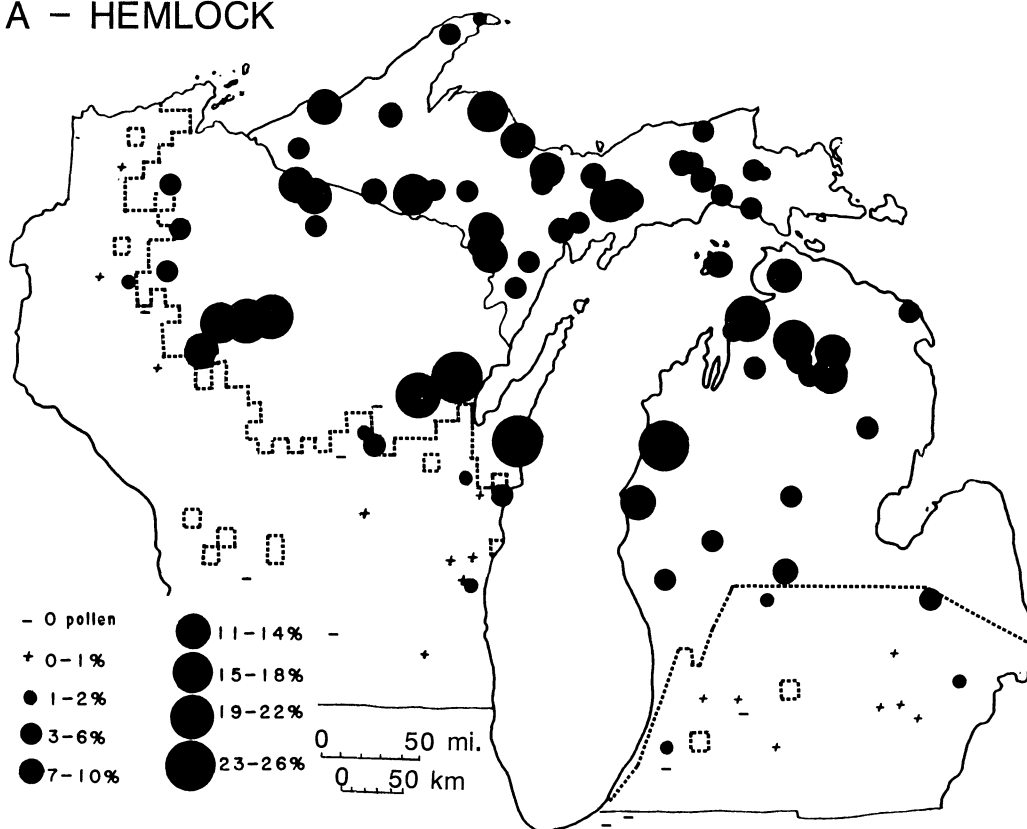
Even a species boundary generalized to township will not include every single tree, however. In Dickinson County, Michigan, beech has been observed 10 km west of the westernmost township where beech was recorded by the

Land Survey. Other isolated occurrences of single trees occur a few kilometres west of the limit suggested by our map (K. D. Woods, unpubl. data). We conclude that it is impossible to map the presettlement species limit or the limit of living trees with absolute precision; it would be unreasonable to expect fossil data to provide information that is any more precise, that is, that could locate the species limit for beech or hemlock in the past more precisely than within 10 km.

Definition of local presence

‘Local’ presence of a tree species is defined in our study to mean occurrence within the township where the lake collecting pollen was located. A township is 36 miles² (92.2 km²), equivalent in size to a circular area of 5.4 km radius.

A – HEMLOCK



B – BEECH

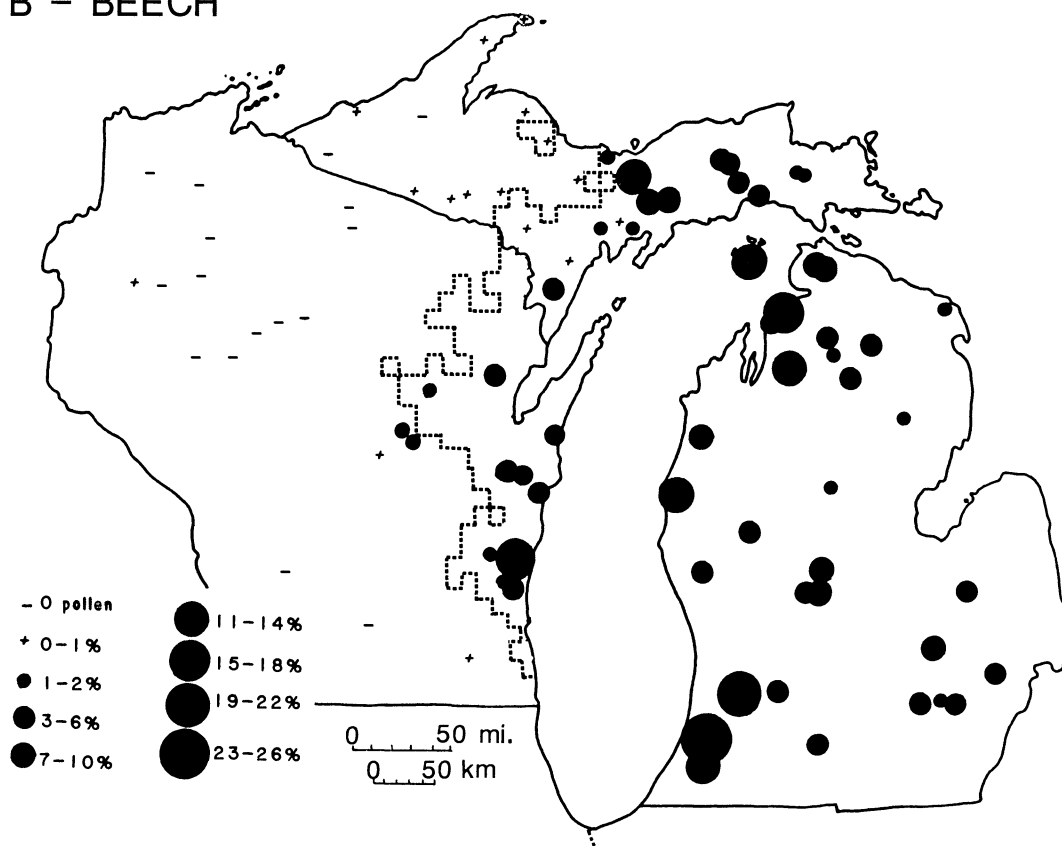


FIG. 4. Dot map of presettlement pollen percentages for (A) hemlock (*Tsuga canadensis*) and (B) beech (*Fagus grandifolia*). Generalized range limits are shown.

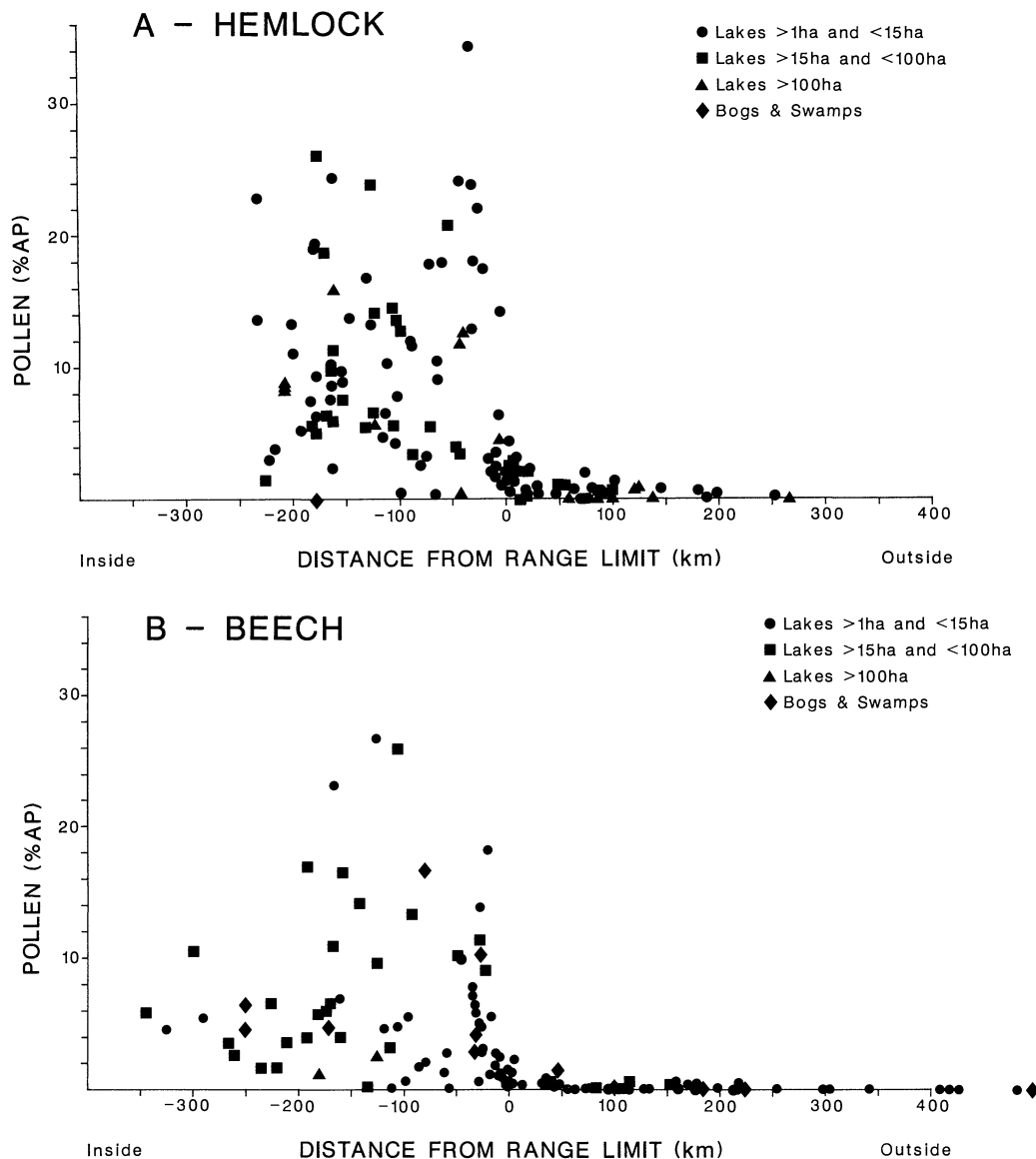


FIG. 5. Scatter plot of hemlock (*Tsuga canadensis*) and beech (*Fagus grandifolia*) pollen percentages against distances of sites from the generalized species range limits.

This definition is based on the arguments outlined above, which led us to generalize range maps to township. In addition, hemlock and beech pollen percentages in small (3–10 ha) lakes correlated better with frequencies of trees within a single township than with trees in a larger area (Schwartz, 1989). A still smaller area may provide most of the pollen, but our studies are not sufficiently precise to analyse data at that scale. Bradshaw & Webb (1985) reach the similar conclusion that most hemlock pollen captured in ‘small and moderate-sized’ lakes (<150 ha) comes from <30 km distance, and beech pollen from substantially shorter distances (Bradshaw & Webb, 1985).

Bennett (1983) defines ‘local’ as growing within the catchment. Several authors studying pollen in surface sediments have used the term ‘local’ for plants growing directly over the coring site (Janssen, 1966, 1981; Davis, Brubaker

& Beiswenger, 1971). Most authors discussing the ‘arrival’ of migrating species, however, imply that they are referring to arrival within an area that extends 5–10 km in all directions. A still larger area, perhaps extending 50–100 km in all directions, is implied by the term ‘regional arrival’ (Davis, 1983). Our township definition of ‘local’ arrival seems compatible with the meaning implied by much of the literature.

Assessing the geographical method for detecting the range limit

In order to see how pollen percentages relate to geographical ranges of trees, we plotted pollen percentages on a map of site locations (Figs. 4A and 4B). Percentages of hemlock and beech pollen were higher in lakes within the

species limits than they were in those outside the limit. Percentages as high as 3% were found immediately outside the limit, but beyond 50–100 km only trace quantities of pollen were found, and at most sites pollen was completely absent.

To inspect the relationship more closely, we plotted hemlock and beech pollen percentages against the distances of the lakes from the generalized boundaries (Fig. 5). Hemlock percentages varied from 1% to 28.4% within the geographical range of hemlock in Wisconsin and Michigan. Outside the range, however, they fell abruptly. Percentages varied between 0 and 4.2% within the first 15 km, but beyond 20 km percentages were consistently lower than 1.0%. Many zero values were found, especially at sites more than 100 km away from the range limit (Table 1). Beech percentages (Fig. 4B) decreased even more abruptly. Percentages as high as 1.9% occurred 15 km outside the boundary, but at more distant sites values were consistently <0.5% (except Volo Bog, discussed below), and many zero values occurred.

Assessing the regression method

Presettlement pollen percentages from thirty small (3–10 ha) Wisconsin lakes were plotted against the percentages of bearing trees in the surrounding townships (Schwartz, 1989). Significant positive straight-line relationships were noted for hemlock and beech pollen versus trees. The slopes of the regression lines were different for hemlock and beech, reflecting differences in pollen representation (Schwartz 1989). Positive *y*-intercept for hemlock (0.6%) suggested that small amounts of ‘background’ pollen came from beyond the 5.4 km radius of the forest township, but the value of the intercept was influenced by the large proportion of sites with zero trees (i.e. located outside the range limit). The *y*-intercept for beech was close to zero (0.08%). Confidence limits were calculated (a) using the variance of the regression line, and (b) calculating the standard error of pollen percentages at all sites with zero trees. Intercepts and confidence intervals are listed in Table 2.

TABLE 2. Expected pollen percentage (*y*-intercept) for beech and hemlock when tree frequency equals zero within the sampling radius of 5.4 km. For beech and hemlock, the first C.I. is based on the error of the regression intercept, using 31 small lakes in Wisconsin. The second C.I. for beech is based on the error of regression, but excludes all sites more than 100 km from the beech range limit. The second C.I. for hemlock is based on the standard error of pollen percentages at all sites where hemlock tree frequencies are zero (Schwartz, 1985).

Species	Expected value	(95% C.I.)	<i>n</i>
Beech	0.08	(−0.01, 0.9)	31
	0.24	(−0.02, 2.2)	16
Hemlock	0.6	(−1.6, 2.8)	31
	1.0	(−0.3, 4.7)	12

DISCUSSION AND CONCLUSIONS

Comparison of geographical and regression methods

The geographical and regression methods provided similar estimates of ‘critical values’. The *y*-intercept for hemlock, representing background pollen, was 0.6%; this value is within the range of pollen percentages, 0–1%, found at sites more than 20 km outside the hemlock range limit. The lower value of the *y*-intercept for beech, 0.08%, corresponds with the low pollen percentages, 0–0.5%, found at sites >15 km outside the beech limit. Thus the two methods identified similar ‘critical’ percentage values, and each method identified a higher critical value for hemlock than beech.

The two methods differ, however, in the precision they provide for distinguishing among the following situations: (a) source trees present locally as part of a continuous population, (b) source trees present locally as a small, outlying population, (c) source trees absent locally and regionally. Presence or absence of a continuous population (a, c) is critical to paleoecologists reconstructing communities, or to those using range limits and thresholds of temperature tolerance as evidence for paleoclimate. Presence or absence of outlying populations (b, c) is a critical question to ecologists considering mechanisms of dispersal, invasion of communities and the dynamics of population growth. It is therefore very important to assess the certainty with which local presence or absence, and continuous versus disjunct populations can be distinguished from fossil pollen evidence.

The geographical method for estimating critical values relies on a grid of sites 25–50 km apart extending across the range limit. A similar network of fossil sites will therefore be necessary to apply geographical information to determine the location of the past range boundary. Our data show clearly that sites within the species’ limit present a range of pollen values, depending on the local density of the source tree. Many fossil sites are necessary to identify the location of the past range limit, because one must observe three tiers of adjacent sites: a tier with very low values that characterize sites more than 20 km beyond the range limit, a tier or band with intermediate pollen values that characterize sites within 20 km of the range limit, and a third series of sites that include the high percentage values found only within the range.

For example, the hemlock range boundary in Michigan–Wisconsin 5000 yr BP was indicated by a series of adjacent sites where fossil pollen values ranged from 1% to 4% bordered on one side (outside the paleo-boundary) by sites with 0–1%, and on the other (inside the paleo-boundary) by sites with higher values. The location of the beech species boundary at that time was indicated by a series of adjacent sites where fossil pollen values ranged from 0.5% to 2%, bordered on one side (outside the paleo-boundary) by sites with 0–0.5% pollen, and on the other side (inside the paleo-boundary) by sites with higher values. This method has been used to trace the invasion of the lake states by hemlock and beech during the Holocene (Davis, 1987).

The range limits for beech or hemlock cannot be defined precisely at any one location, however. Instead they are located within a band of sites 15–20 km wide. Beyond that band beech or hemlock is absent, and inside the band local populations are present; within the band itself local populations may or may not be present. This level of precision is adequate for most purposes, and is only slightly less precise than a map of modern range limits using living trees. Only rarely are fossil sites spaced closer than 20 km on the landscape; the density of fossil sites probably represents the main limit on the precision with which past range limits can be detected.

Regressions are much less powerful for this kind of analysis, because they rely on the identification of a single critical value, the *y*-intercept. The problems are (1) that the *y*-intercept is sensitive to the particular data set used for its calculation, as well as the method of regression (Schwartz, 1985), and (2) that the confidence interval is so wide that for hemlock the *y*-intercept includes pollen percentages as low as 0 and as high as 2.5 or 4.7 (depending on the method of calculating confidence limits). This means that using this method a sediment sample with 4% pollen cannot be considered definitive proof of the *presence* of hemlock trees. Nor can one argue that the *absence* of hemlock is demonstrated by zero pollen values, because the confidence interval for an estimate of pollen, given 1% trees, includes zero values. This is true for beech as well (Schwartz, 1985).

A problem inherent in regression methods is that regressions give equal weight to all sites where tree abundance equals zero. These sites are not all equal, however, as some are close to the locations of trees, while others are hundreds of kilometres distant. The geographical approach, on the other hand, takes dispersal of pollen as well as production into account, giving the analysis greater power.

The problem of identifying small, outlying populations is more difficult than determining local presence of a large, continuous population. Critical values cannot identify small populations because pollen numbers at a single point in space may be identical, whether they originate from a few trees within 5 km or from a large population 100 km distant. An outlier can only be identified from percentages greater for a period of time (represented in several samples) than at surrounding sites. A dense geographical grid is necessary to detect this circumstance; high pollen counts or multiple sites/cores are necessary to overcome statistical error.

In general, our pollen counts (Table 1) are not sufficiently high to identify outliers successfully. Note, however, the high (1%) beech pollen percentages at Volo Bog (King, 1981), a site in northern Illinois within a county where beech trees grow today (Mohlenbrock, 1975). Hemlock pollen percentages are also unusually high at Radtke Lake in southeastern Wisconsin (S. L. Webb, 1987) where the Land Survey tree map indicates a nearby outlying population of hemlock (Fig. 4A).

Outlying colonies close to the continuous range limit may be difficult to detect because of the blurring effect of dispersal from the nearby large population. For example, a core was analysed from a lake within a large outlier of hemlock 15 km west of the species boundary in northwestern Wisconsin (Fig. 1A). The outlying population was too close to the boundary to be distinguished by pollen: hemlock per-

centages were slightly more than 1%, similar to values from adjacent sites where hemlock pollen was windblown from the main population. Similarly, the outlying population itself appeared as a 'peninsula' extending from the main population when the range boundary was generalized to township (Fig. 4A). Similarly equivocal results were obtained in sites from outlying beech populations in northern Michigan. Pollen percentages ranging between 0.5% and 1% occurred at Second Bass Lake, a small pond within a beech outlier, 40 km beyond the main distribution limit. Similar percentages were found at Canyon Lake, 7 km beyond Second Bass and well outside the distribution limit, and at Abraham Lake, which lies at the range limit (Fig. 3). Beech pollen percentages at all three sites fall within the range of values characteristic of sites along the range boundary. Woods & Davis (1989), who analysed these sites with the aim of dating the establishment of beech populations at the westernmost species limit, noted that pollen abundances were increasing at all three sites over the last millennium; it was difficult, however, to pinpoint stratigraphic levels that unequivocally represented local colonization events. Much higher pollen counts (several thousand grains per sample) or a denser sampling grid (one lake every 10 km) might have helped to reduce these ambiguities.

We conclude that it is possible, under favourable circumstances, to use pollen percentages to identify small outlying colonies. Even very low percentages (<1%), if they occur as a result of large pollen counts, and occur consistently in a series of samples or at a series of stratigraphic levels (effectively increasing the sample size) may represent evidence of local trees. However, they must be significantly higher than pollen percentages at surrounding sites, and there must be evidence that no large (area >1000 km²) population exists nearby to serve as a source for windblown pollen. The critical distance from a large pollen source depends on the pollen percentages within the proposed outlier relative to the background levels of pollen dispersed from the large source.

Outlying colonies can easily go undetected, even with a closely-spaced grid of fossil sites, because pollen production by populations of this size may not be large. Most beech and hemlock pollen grains are dispersed only a few kilometres (Bradshaw & Webb, 1985; Schwartz, 1989); the high between-site variance appears to reflect township-scale variations in abundance (compare Figs. 2 and 5). Given short dispersal distances, even a grid of sites 10 km apart could fail to collect sufficient pollen from small populations occupying fewer than 10 km².

GENERALITY OF RESULTS

Using maps of hemlock and beech from a time when the landscape was forested, and pollen in sediment of similar age, we have been able to define criteria by which fossil pollen in sediments can be used to detect the distribution limits for these two tree species. Of course the specific criteria will not apply to other species, nor even to hemlock and beech forests with different pollen productivities.

Hemlock and beech are conducive to a study such as this one for two reasons. First, both species are good pollen pro-

ducers with moderate to low pollen dispersal. Tree species that produce or disperse fewer pollen grains (e.g. *Tilia*) might have more useful critical values for detecting local presence; prolific pollen producers with well-dispersed pollen (e.g. *Pinus*), on the other hand, will show a less distinct pollen decline across the species boundary. A second factor in favour of hemlock and beech is their habit of growing densely within a few kilometres of their absolute distribution limits instead of declining to rarity as do many trees (Brown, 1984). The sharp gradient in density, as well as poor pollen dispersal, is responsible for the marked change in pollen percentages at the species limit. If either tree had been rare near its distribution limit in the past, then the pollen record of its past distribution limits would be less clear. This possibility has been investigated by plotting fossil hemlock and beech pollen percentages from Michigan and Wisconsin lakes along transects perpendicular to the direction of migration (Davis, 1987). The data show that the sharp change in pollen abundance across the species boundaries occurred throughout the Holocene, even as the species' boundaries changed locations. Only during the first millennium of invasion was pollen production too low to provide clear evidence for the location of the species limit (Davis, 1987). Hemlock and beech are unusual; the species limit would be much more difficult to detect for a species like red oak (*Quercus rubra* du Roi), which is rare throughout a several hundred km wide region along its species limit in northern Michigan. The tree grows densely only on dry soils, often where fires have disturbed the site. Because its pollen is well dispersed, the pattern of pollen percentages in surface lake sediments appears more uniform than the pattern of trees. Geographical studies for other tree species similar to our work on hemlock and beech are needed before general statements can be made about the precision with which species limits can be detected in the past. With a network of fossil sites it may then be possible to obtain new information on responses to climatic change and invasions of forest communities.

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